SWE-bench Science: Can Coding Agents Resolve Engineering Tasks in Science?

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Abstract

Software increasingly functions as part of the scientific instrument itself, making failures in scientific code capable of compromising not only program behavior but also the evidence underlying scientific conclusions. Yet existing evaluations of coding agents largely emphasize aggregate task success, providing limited insight into why agents fail when repairing scientific software. We introduce **SWE-bench Science**, a repository-level benchmark for scientific software engineering comprising 119 tasks from 98 GitHub repositories across 20 scientific domains. Each task is organized into one of three paradigms: Issue-driven, Expert-exploratory, and Engineering-integration. Even the best-performing agent, **Claude Code with Opus-5 (max), achieves a pass@1 below 50%**, highlighting the substantial challenges posed by scientific software engineering. We identify four recurring failure mechanisms: deficits in scientific knowledge or abstraction, misguided exploration or surface-level repair, incomplete repair coverage or system integration, and failures to generalize scientific knowledge beyond observed cases in our analysis. We further conduct a paired ablation that removes explicit scientific guidance while preserving the repository and executable engineering context. The results show that scientific knowledge is not uniformly beneficial: well-grounded information can constrain repair and improve average performance and token efficiency, whereas poorly aligned guidance can induce anchoring and does not necessarily improve exact repair success. Together, SWE-bench Science provides a broad testbed for studying both the capabilities and failure mechanisms of coding agents in scientific software engineering.

Code: https://github.com/OpenMOSS/SWE-bench-Science
Data: https://huggingface.co/datasets/OpenMOSS-Team/SWE-bench-Science
Leaderboard: https://swescience.github.io

1 Introduction

Software is no longer merely an aid to science; it is part of the instrument through which scientific claims are produced. Research groups rely on code to run simulations, process instrument outputs, train surrogate models, search chemical or biological candidates, manage high-throughput experiments, and reproduce published results. A defective patch can therefore corrupt not only a program output but also the evidence behind a scientific conclusion.

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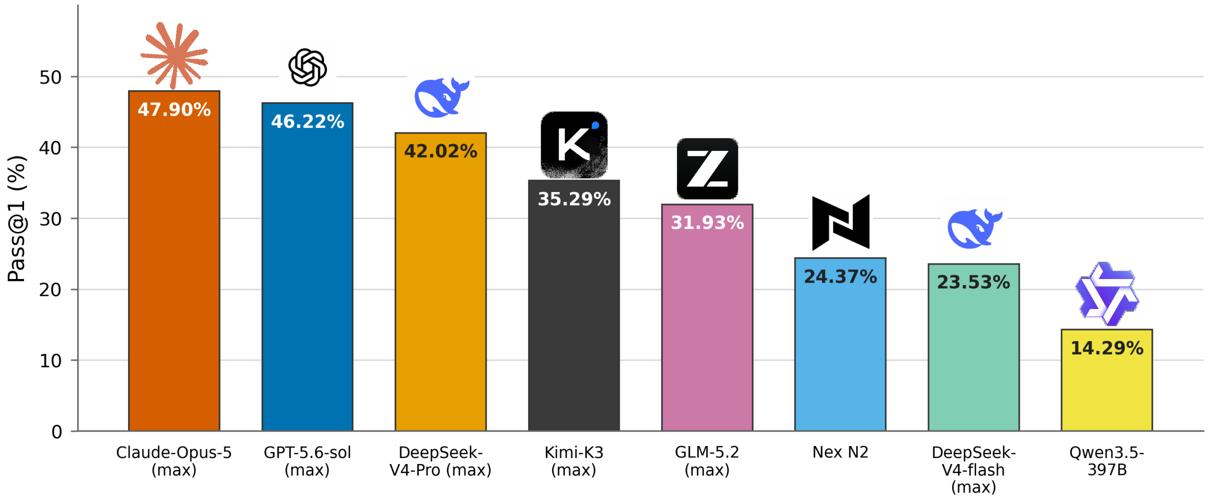


Figure 1 Pass@1 comparison of coding agents on SWE-bench Science.

Understanding why coding agents fail on scientific software is as important as measuring whether their patches pass tests. A failed repair may reflect an incorrect scientific abstraction, superficial exploration of an observed symptom, incomplete integration across a software system, or an inability to generalize a scientific principle beyond visible cases. Aggregate test scores cannot distinguish these sources and therefore provide limited guidance for improving scientific coding agents. Existing benchmarks measure function synthesis and repository-level software repair [1–3], while scientific coding benchmarks cover curated research problems, paper-grounded implementations, scientific workflows, repository execution, and production scientific codebases [4–9]. However, repository-level coverage across scientific domains and the analysis of failure mechanisms remain limited, leaving broad cross-domain scientific software engineering underexplored.

We introduce **SWE-bench Science**, a repository-level benchmark containing 119 tasks from 98 unique GitHub repositories and spanning 20 scientific domains. Each task is manually inspected to verify its scientific contract, reproducibility, and evaluation validity. We organize the benchmark into three scientific task paradigms—**Issue-driven**, **Expert-exploratory**, and **Engineering-integration**—that capture complementary requirements in scientific software maintenance. Beyond measuring task success, we manually analyze unsuccessful attempts through four recurring scientific failure mechanisms: scientific-knowledge or abstraction deficits, misguided exploration or surface-level repair, incomplete repair coverage or system integration, and failures of scientific-knowledge generalization. Runtime or evaluation-path failures are recorded separately. We further construct a paired comparison that removes explicit scientific auxiliary information while preserving the repository and executable engineering context. The resulting analysis shows that scientific knowledge is not uniformly beneficial: its effect depends on how the knowledge is organized and grounded in executable evidence, because well-targeted information can constrain the repair while poorly aligned information can induce anchoring or substitute for independent validation.

Our contributions are:

1. We introduce a broad-coverage scientific software engineering benchmark with 119 tasks from 98 unique GitHub repositories, spanning 20 scientific domains, and evaluate representative frontier coding agents on these tasks.

2. We analyze unsuccessful scientific software repairs and identify four recurring failure mechanisms, revealing concrete directions for improving scientific abstraction, disciplined exploration, system-wide repair, and scientific-knowledge generalization.

3. We provide a paired scientific-knowledge ablation that removes explicit scientific guidance while preserving the executable engineering context. The results show that the organization and grounding of scientific

knowledge matter in scientific software engineering: additional information can improve average scores and token efficiency, but does not automatically improve exact repair success.

2 Related Work

2.1 Code and Software Engineering Benchmarks

Early code-generation benchmarks evaluate whether models can synthesize short functions from natural-language prompts. HumanEval [1] and MBPP [2] are representative examples, focusing on unit-test correctness for self-contained programming problems. Later benchmarks increase realism by requiring repository-level modification and issue resolution. SWE-bench [3] evaluates whether language models can resolve real GitHub issues by generating patches against project test suites. More recent benchmarks extend this setting toward longer-horizon agentic engineering: SWE-Bench Pro targets complex enterprise-level repository tasks, DeepSWE uses original tasks and hand-written verifiers to reduce contamination and oracle ambiguity, and Terminal-Bench evaluates agents on hard, realistic tasks in command-line environments [10–12]. Together, these benchmarks measure increasingly realistic software engineering capabilities, including repository inspection, fault localization, multi-file modification, tool use, and test-validated repair.

2.2 LLMs for Scientific Workflows

Scientific-code benchmarks mainly study two settings. SciCode and MMSciCode evaluate self-contained scientific programming and paper-grounded function implementation, respectively, making them closer to non-agentic programming evaluation than to repository-level software engineering [4, 5]. A second line evaluates coding within broader research workflows: ResearchCodeBench and LMR-BENCH study research-paper code implementation and research-code reproduction [13, 14]; ScienceAgentBench, SUPER, and CSR-Bench emphasize data-driven scientific discovery, research-repository execution, and research-system deployment [6, 7, 15]; and RExBench, AutoMat, and AstaBench evaluate research-code extension, scientific workflow recovery, and multi-stage scientific tasks [8, 16, 17]. AInsteinBench is closest to our setting because it targets scientific computing through issue-derived and synthesized feature tasks in production repositories [9]. SWE-Bench 5G studies specification-dependent telecom repair [18]. SWE-bench Science instead centers scientific software engineering as an applied engineering problem: coding agents must inspect production repositories, modify interacting components, and preserve scientific validity across **Issue-driven**, **Expert-exploratory**, and **Engineering-integration** tasks. This focus differs from solving standalone scientific programming problems or completing broader research workflows by evaluating the maintenance and extension of scientifically valid software systems.

Benchmark	Task count	# Scientific domains	GitHub repos	Scientific focus	Task source
SciCode [4]	80	16	–	Function programming	expert
MMSciCode [5]	624	6	–	Function programming	expert
ResearchCodeBench [13]	212	1	20	Research code	expert
LMR-BENCH [14]	28	1	23	Research code	expert
ScienceAgentBench [6]	102	4	30	Research code	expert
SUPER [7]	801	–	694	Research code	expert
CSR-Bench [15]	100	5	100	Research code	–
RExBench [16]	12	–	–	Research code	expert
AutoMat [17]	85	1	–	Research code	expert
AstaBench [8]	>2,400	–	–	Research code	expert
AInsteinBench [9]	244	6	6	Scientific computing	issue/expert
SWE-Bench 5G [18]	210	1	3	Scientific software engineering	issue
SWE-bench Science (ours)	119	20	98	Scientific software engineering	issue/expert/engineer

Table 1 Comparison of coding-agent benchmarks for science. SWE-bench Science covers 20 scientific domains.

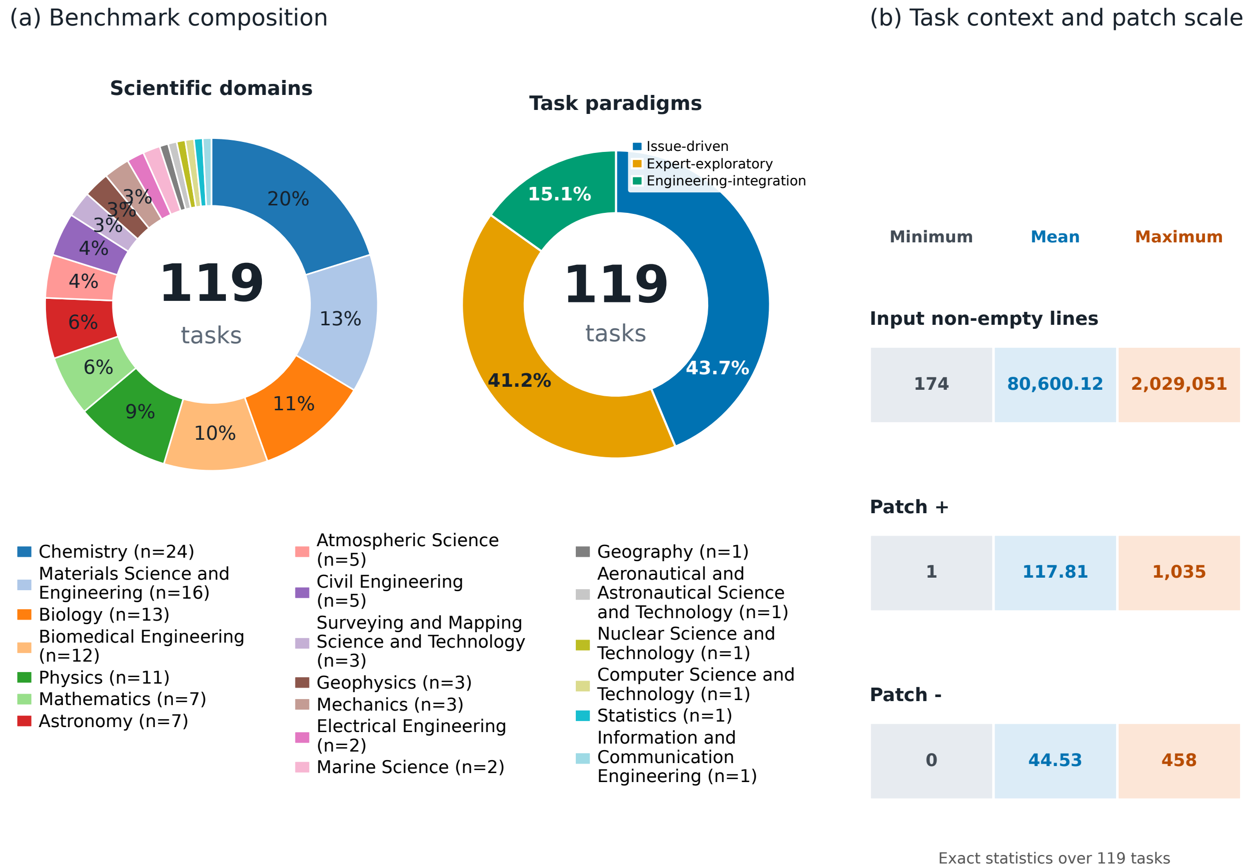


Figure 2 Benchmark coverage and scale over 119 tasks. (a) Scientific-domain distribution and task-paradigm composition. The domain legend reports the task count for each of the 20 domains; the task-paradigm chart contains 52 Issue-driven tasks (43.7%), 49 Expert-exploratory tasks (41.2%), and 18 Engineering-integration tasks (15.1%). Chemistry is the largest domain with 24 tasks, followed by Materials Science and Engineering with 16, Biology with 13, Biomedical Engineering with 12, and Physics with 11. (b) Minimum, mean, and maximum input and reference-patch sizes, shown directly in horizontal summary bands. Input size is measured as non-empty physical lines in the task code, while Patch + and Patch − count additions and deletions in the standard reference diff.

3 SWE-bench Science

We cover 98 unique GitHub repositories and construct 119 tasks across 20 scientific domains, with the domain and task-paradigm composition shown in Figure 2(a). The five largest domains contain 76 tasks (63.9%), while six domains contribute one task each. The task set contains 52 Issue-driven, 49 Expert-exploratory, and 18 Engineering-integration tasks. The complete scientific-domain counts are reported in Appendix Table 4. The benchmark also spans substantial variation in repository context and repair size. Figure 2(b) shows that non-empty input code averages 80,600.12 lines and ranges from 174 to 2,029,051 lines. Reference patches add 117.81 lines on average, with a range of 1–1,035, and delete 44.53 lines on average, with a range of 0–458. Direct labels preserve the exact values despite the large scale differences among these quantities.

3.1 Task Schema

The following fields are visible to the agent in the standard condition:

- **Repository snapshot**: the exact pre-task state with locked dependencies and runnable entry points, stripped of Git history, remotes, future changelogs, build caches, and solution-linked artifacts.

- **Problem statement**: a specification frozen before test and reference-patch authors inspect one another's artifacts.
- **Required scientific context** c_i^{req}: local, versioned definitions and constraints required for a well-posed task; it is held fixed in every condition.
- **Public tests**: visible software and scientific assertions that support interactive debugging.

Evaluator-only fields include private tests, contract labels, reference and alternative-valid patches, scientific-rationale and localization support blocks, expected files, source and difficulty metadata, contamination tier, and post-hoc annotations. Private tests are mounted only after patch submission in a separate evaluator container. None of fields is available through the agent workspace, environment variables, logs, or task metadata.

3.2 Example.

Figure 3 illustrates an evaluation pipeline in the benchmark for an example task. The task asks the agent to reconcile two periodic representations of the same crystal while preserving their physical energy. The example shows the agent-visible problem statement, scientific context, and workspace snapshot as the input to the agent-driven coding loop and the clean evaluation phase. Public checks support debugging, whereas private scientific cases determine whether the submitted patch completes the intended repair.

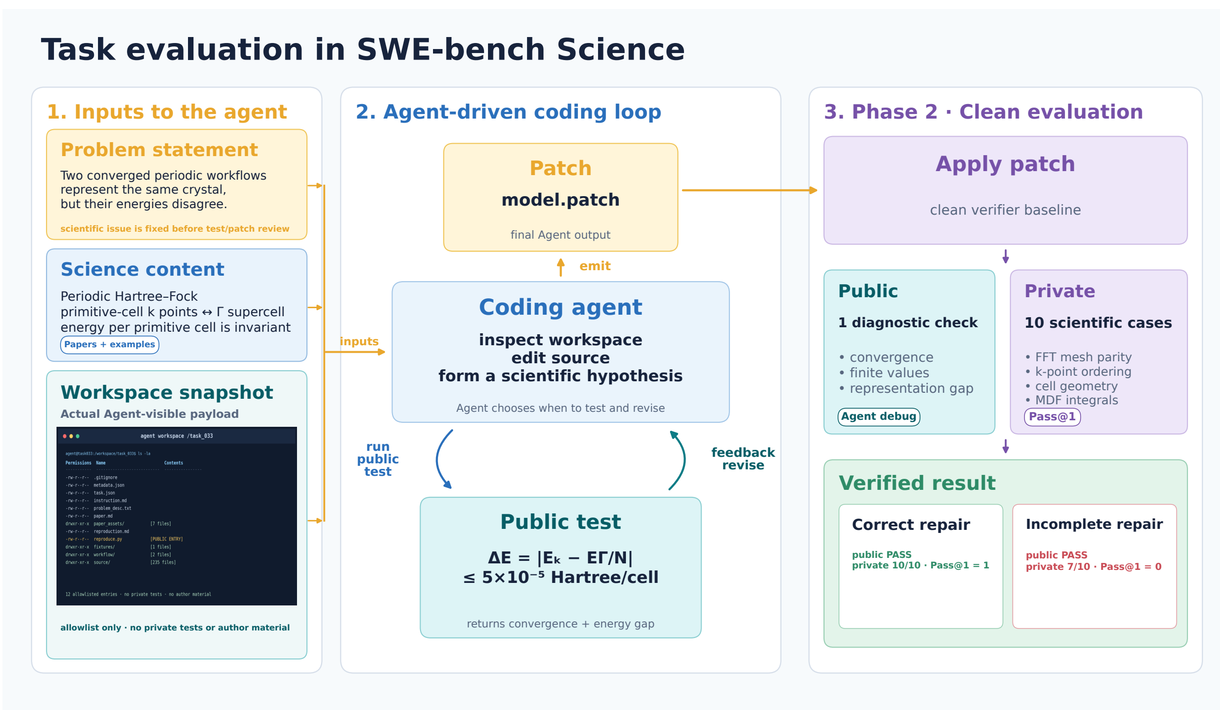


Figure 3 Evaluation pipeline in SWE-bench Science for an example task. The diagram shows the frozen agent-visible inputs, the agent-driven coding loop, and the clean evaluation phase with separate public diagnostics and private scientific cases.

4 Benchmark Construction

To evaluate agent performance across different capability dimensions in real scientific software development and maintenance, we establish a unified Chain-of-Evidence Protocol and derive three task paradigms with clearly differentiated evaluation goals: **Issue-driven**, **Expert-exploratory**, and **Engineering-integration** tasks. Issue-driven tasks emphasize localized repair and regression avoidance, Expert-exploratory tasks require autonomous investigation of scientific discrepancies, and Engineering-integration tasks evaluate cross-module completion of end-to-end scientific workflows.

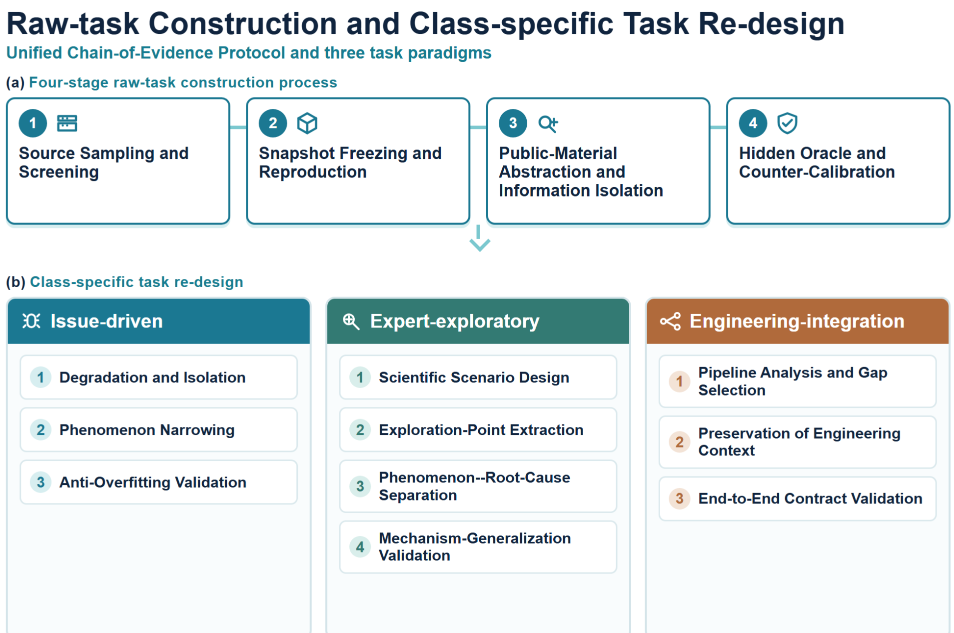


Figure 4 Combined overview of raw-task construction and class-specific task re-design.

All raw tasks are first generated according to a unified and auditable four-stage construction process. The process is designed to control environmental interference and prevent information leakage.

1. **Source Sampling and Screening.**

   We collect real issues, pull requests (PRs), commit records, and related literature from candidate open-source scientific-computing repositories. We then apply predefined screening criteria to remove overly simple fixes, tasks with unstable environment dependencies, tasks with severe solution leakage, and projects that substantially overlap with existing samples.

2. **Snapshot Freezing and Reproduction.**

   We freeze the source-code snapshot immediately before the target defect or missing capability occurred, denoted by $\mathcal{S}_{\text{bug}}$. The snapshot is executed in an isolated container to verify that the target anomaly or capability gap can be reproduced reliably. This step ensures that the observed failure is caused by the core algorithm or logical semantics rather than by environmental noise.

3. **Public-Material Abstraction and Information Isolation.**

   Based on the intended behavioral contract, we extract the scientific invariants and data-flow constraints evaluated by each task. We construct a public evaluation package containing the public repository, a coarse-grained description of the observed phenomenon, a reproduction script, and the necessary domain-background documentation. The public materials do not reveal the patch location or hidden assertions.

4. **Hidden Oracle and Counter-Calibration.**

   We construct hidden validation suites based on semantic equivalence and boundary conditions. The validators examine not only the execution results under normal inputs, but also include reverse checks for hard-coded solutions, heuristic pseudo-fixes, and incomplete repairs.

Raw tasks are further re-designed according to their fit with the three task paradigms. Figure 4 summarizes the four-stage raw-task construction process and the class-specific task re-design for the three paradigms.

4.1 Class-Specific Task Re-design

Through the differentiated construction strategies described below, we re-design the raw tasks to enable the benchmarking of three capabilities:

- **Issue-driven tasks** focus on repairing known defects;
- **Expert-exploratory tasks** focus on autonomously reasoning about unknown root causes in scientific scenarios;
- **Engineering-integration tasks** focus on understanding architecture across files and connecting a complete multi-module capability chain.

This task taxonomy avoids conflating different agent capabilities and provides a structured experimental basis for evaluating large language models in realistic scientific software engineering settings.

4.1.1 Issue-Driven Tasks

Definition and paradigm. Issue-driven tasks evaluate an agent's ability to repair real bugs under low-distortion conditions. These tasks take a confirmed historical defect as their direct definition starting point. Their central design principle is to narrow the observable phenomenon while isolating the patch information.

Construction process.

1. **Degradation and Isolation.**

   We analyze the historical issue and PR discussion, roll the source code back to the state before the problematic commit, and construct a minimal reproducible example (MRE).

2. **Phenomenon Narrowing.**

   We reduce the upstream error to a coarse-grained and auditable phenomenon, such as inconsistent results across different computational paths or data loss under a particular format. The public materials remove any specific hints about the solution patch.

3. **Anti-Overfitting Validation.**

   Hidden validators cover different data scales, boundary parameters, and input-order transformations. These validators test whether the repaired code genuinely restores the scientific semantics rather than merely fitting the public script.

4.1.2 Expert-Exploratory Tasks

Definition and paradigm. Unlike Issue-driven tasks, which directly reproduce a known defect, Expert-exploratory tasks simulate a real scientific exploration process. Their definition starts from an exploration space containing a scientific scenario. Some tasks may be inspired by historical issues or PRs, but those issues or PRs do not serve as the definition starting point. These tasks evaluate how an agent uses domain knowledge to perform autonomous black-box or gray-box reasoning when the root cause of a complex phenomenon is unknown.

Construction process.

1. **Scientific Scenario Design.**

   We first define a high-level scientific use case, such as molecular representation consistency calibration, measurement-chain bias analysis, medical-image coordinate alignment, or experimental-workflow verification.

2. **Exploration-Point Extraction.**

   From theoretical materials, methodological literature, experimental phenomena, or discrepancies between results, we identify a core difference that is worth actively exploring. Upstream issues or PRs may provide optional background evidence, but they do not define the task.

3. **Phenomenon–Root-Cause Separation.**

   We construct a runnable public workflow that preserves the relevant domain background and observable anomaly, such as abnormal convergence or biased results, while hiding the precise location of the defective source code. The agent must identify the root cause through autonomous observation, controlled comparisons, and scientific reasoning.

4. **Mechanism-Generalization Validation.**

   Hidden validators change parameter scales, physical topologies, coordinate orderings, or boundary conditions. These tests determine whether the agent has genuinely inferred the scientific mechanism behind the observed phenomenon.

4.1.3 Engineering-Integration Tasks

Definition and paradigm. Engineering-integration tasks go beyond single-point bug repair. They focus on evaluating an agent's ability to understand system architecture in a realistic and complex codebase, connect different modules, and complete a full capability chain.

Construction process.

1. **Pipeline Analysis and Gap Selection.**

   We analyze the repository's end-to-end call chain, including data loading, parameter interpretation, intermediate-representation (IR) construction, operator assembly, and numerical solving. We then select a functionality gap with an appropriate scope.

2. **Preservation of Engineering Context.**

   The public source code preserves the complete real package structure and neighboring modules. Public reproduction scripts show state comparisons at multiple stages and identify where the functionality chain fails to connect correctly. This design requires the agent to navigate across files and develop a system-level understanding.

3. **End-to-End Contract Validation.**

   Hidden validators include alternative execution paths, state-reset tests, and inter-module behavioral-contract tests. These validators ensure that the proposed repair achieves architecture-level integration across the entire module capability chain rather than merely making a single public test pass.

4.2 Scientific Auxiliary Information Separation

The scientific-information factor measures the contribution of externally supplied domain knowledge, instantiated as scientific-rationale support. Of the 119 tasks, 91 allow this information to be separated from the supplied materials. For each eligible task, we compare paired conditions that differ only in auxiliary scientific information; the source-code snapshot, execution environment, public reproduction entry points, hidden validators, and required scientific context c_i^{req} remain fixed. Scientific auxiliary information denotes evidence relevant to scientific validity that is not directly available from source code or observable runtime behavior, including scientific rationales, upstream repairs, audit findings, paper excerpts, and expert guidance.

Withholding this information does not remove repository-intrinsic engineering cues, such as code structure, interfaces, traces, tests, and incomplete implementations, because doing so would change the software task itself. The ablation retains the executable context, task objective, observable symptoms, input data, and minimal interface documentation while removing scientific principles, equations or assumptions, expected properties, domain-specific diagnoses, and scientifically motivated repair strategies. It therefore estimates the marginal contribution of scientific auxiliary information given the same repository evidence, rather than performance in a clue-free setting. Results are analyzed in Section 6.2.

5 Experiments

Coding Agents. We evaluate eight agent configurations in our benchmark: GPT-5.6-sol with Codex at **max** reasoning depth, Claude-Opus-5 with Claude Code at **max**, Kimi-K3 with Kimi Code at **max**, GLM-5.2 with Codex at **max**, Nex N2 with Codex, DeepSeek-V4-flash-0731 with Claude Code at **max**, Qwen3.5-397B with Codex, and DeepSeek-V4-Pro-0813 with Claude Code at **max**. The comparison is shown in Table 2.

Evaluation. We evaluate each submitted patch with public and private tests while keeping private tests and evaluator-only metadata outside the agent workspace. For each task–attempt, PublicScore is the mean score across all applicable public test cases, and PrivateScore is the corresponding mean across private test cases. Fail2Pass is the fraction of previously failing private tests that pass after the repair, while Pass2Pass is the fraction of previously passing private tests that remain passing; we set Pass2Pass to 1 when the previously passing set is empty. Pass@1 is a binary exact-private-success measure that equals 1 only when every applicable private test passes. Table 2 reports task-level means for the public score, private score, Fail2Pass, Pass2Pass, and overall Pass@1, together with Pass@1 for Issue-driven, Expert-exploratory, and Engineering-integration tasks.

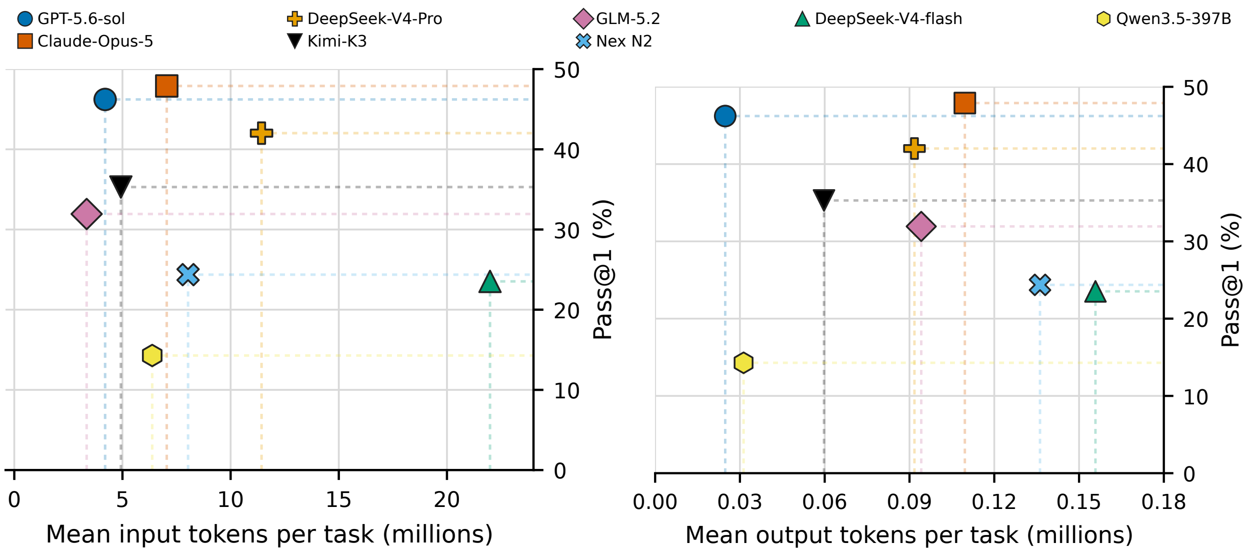
5.1 Performance and Token Consumption

LLM	**Harness**	**Public score**	**Private score**			**Pass@1**			
			Private score	Fail2Pass	Pass2Pass	**Overall**	Issue	Expert	Engineering
GPT-5.6-sol (max)	Codex	99.16%	**78.82%**	**72.30%**	**97.66%**	46.22%	36.54%	59.18%	38.89%
Claude-Opus-5 (max)	Claude Code	96.64%	75.11%	68.60%	97.37%	**47.90%**	**38.46%**	**65.31%**	27.78%
DeepSeek-V4-Pro (max)	Claude Code	**100.00%**	73.16%	65.77%	96.58%	42.02%	26.92%	57.14%	**44.44%**
Kimi-K3 (max)	Kimi Code	98.32%	66.34%	57.55%	94.94%	35.29%	25.00%	44.90%	38.89%
GLM-5.2 (max)	Codex	94.12%	63.61%	53.81%	97.53%	31.93%	17.31%	46.94%	33.33%
Nex N2	Codex	93.28%	61.89%	51.09%	94.92%	24.37%	11.54%	36.73%	27.78%
DeepSeek-V4-flash (max)	Claude Code	98.32%	61.41%	52.34%	95.74%	23.53%	19.23%	26.53%	27.78%
Qwen3.5-397B	Codex	96.64%	51.79%	38.33%	95.16%	14.29%	5.77%	24.49%	11.11%

Table 2 Main experimental results. Scores are task-level means over the common 119 tasks. The best value in each column is bold and underlined; ties are retained.

Table 2 shows that no single model leads every metric. DeepSeek-V4-Pro achieves the best public score and performs best on Engineering-integration tasks. GPT-5.6-sol leads the private score, Fail2Pass, and Pass2Pass, indicating that it is the strongest at both repairing failed private tests and preserving tests that already pass. Claude-Opus-5 achieves the best overall Pass@1 and leads on both Issue-driven and Expert-exploratory tasks. Even frontier models achieve Pass@1 scores below 50%, underscoring the challenge of SWE-bench Science.

Figure 5 further shows the Pass@1 score versus token consumption for those agents. Claude-Opus-5 achieves



(a) Mean input-token consumption.

(b) Mean output-token consumption.

Figure 5 Pass@1 versus mean token consumption per task over the common 119-task evaluation. Dashed guides connect each observation to the corresponding axes.

the highest Pass@1 with a moderate token budget, while GPT-5.6-sol reaches a similar level with much shorter outputs. DeepSeek-V4-Pro uses more input tokens but ranks behind both. Kimi-K3 and GLM-5.2 achieves lower performances but consume fewer input tokens. For smaller models under 400B, Nex-N2 achieves best performance-token balance, better than DeepSeek-V4-flash and Qwen3.5-397B. The main difference therefore lies in how effectively each model–harness configuration uses its context and generation budget, rather than in token volume alone.

6 Analysis

6.1 Observed Failure Mechanisms

LLM	Harness	Total errors	Knowledge/ abstraction	Exploration/ surface repair	Repair coverage/ system integration	Scientific generalization
GPT-5.6-sol (max)	Codex	64	18	10	22	14
Claude-Opus-5 (max)	Claude Code	**58 (+4)**	24	**2**	21	11
DeepSeek-V4-Pro (max)	Claude Code	69	**15**	12	**19**	23
Kimi-K3 (max)	Kimi Code	77	20	14	22	21
GLM-5.2 (max)	Codex	81	20	14	24	23
Nex N2	Codex	90	31	14	23	22
DeepSeek-V4-flash (max)	Claude Code	91	23	14	48	**6**
Qwen3.5-397B	Codex	102	26	15	33	28

Table 3 Error-count breakdown by coding agents over the common 119 tasks. The total column records attempts assigned to the four mutually exclusive scientific failure mechanisms defined in the text; for Claude-Opus-5, the parenthesized +4 denotes additional runtime or evaluation-path failures. The four category counts sum to the main total for each row. The lowest count in each column is bold and underlined.

We use four recurring scientific failure mechanisms in the audit reports. **Scientific-knowledge or abstraction deficit** denotes a repair based on an incorrect or incomplete scientific object, mathematical definition, or domain abstraction. **Misguided exploration or surface-level repair** denotes a patch that addresses the visible symptom or public metric without tracing the failure to an independent oracle or the underlying scientific contract. **Incomplete repair coverage or system integration** denotes a locally plausible repair that does not satisfy the requirements of the full software system: for example, one module is corrected while its interactions, data flow, shared invariants, or compatibility with other modules remain unpreserved. **Failure of scientific-knowledge generalization** denotes a repair that handles the observed scientific case but does not extend the same scientific principle to unseen conditions, equivalent representations, boundary regimes, or other variants that require scientific generalization. Claude-Opus-5 additionally has four **runtime or evaluation-path failures**, in which an attempt does not complete the intended execution or evaluation path and therefore cannot be attributed to one of the four scientific mechanisms.

Claude-Opus-5 produces the lowest categorized scientific-error count (58, plus 4 runtime or evaluation-path failures) and the fewest misguided-exploration or surface-level-repair errors (2). DeepSeek-V4-flash (max) records the fewest scientific-knowledge generalization errors (6), while DeepSeek-V4-Pro records both the fewest scientific-knowledge or abstraction errors (15) and the fewest incomplete-repair or system-integration errors (19).

6.2 How Scientific Information Affects Performance

On the 91-task separable subset, we compare GPT-5.6-sol (xhigh) with DeepSeek-V4-flash (high), each with and without scientific information. Figure 6 summarizes their mean scores and token use.

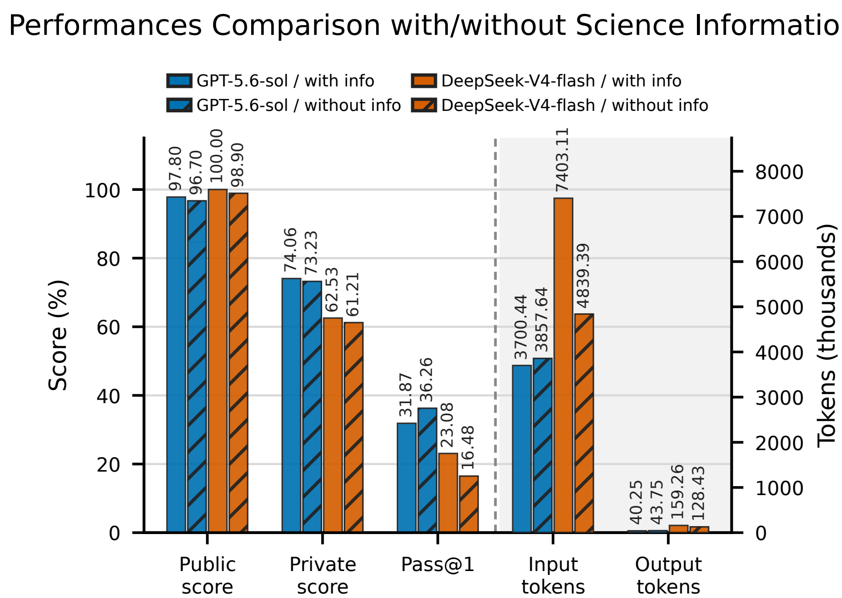
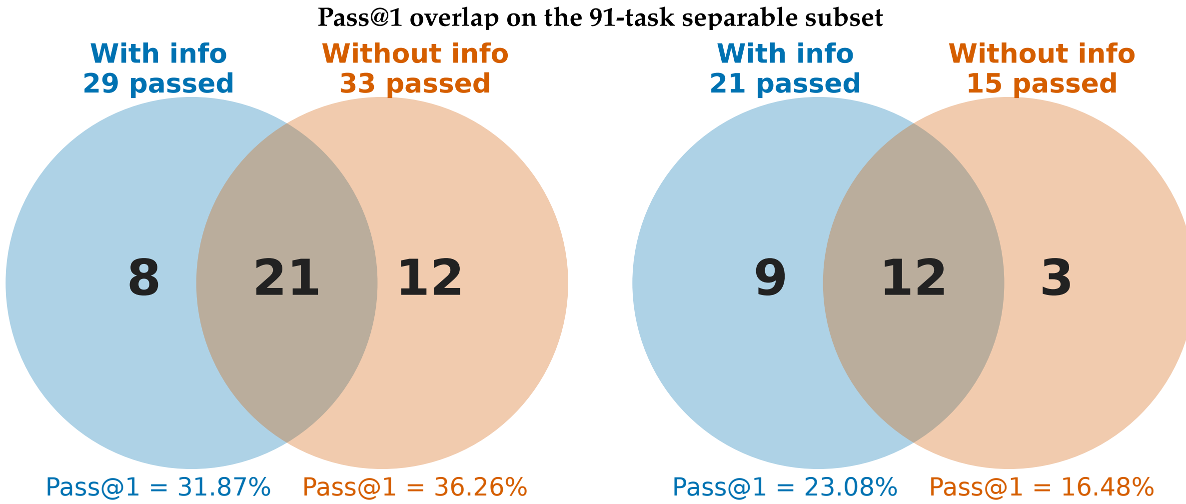


Figure 6 Scores and token consumption over the 91 tasks whose scientific-information content differs between conditions. GPT-5.6-sol and DeepSeek-V4-flash are each shown with and without scientific information on the same task subset.

For GPT-5.6-sol, scientific information slightly increases the mean public and private scores from 96.70% and 73.23% to 97.80% and 74.06%, but decreases Pass@1 from 36.26% to 31.87%. Mean input and output tokens also decrease from 3.86 million and 43.75 thousand to 3.70 million and 40.25 thousand. For DeepSeek-V4-flash, the same intervention raises the public score, private score, and Pass@1 from 98.90%, 61.21%, and 16.48% to 100.00%, 62.53%, and 23.08%, while increasing input and output tokens from 4.84 million and 128.43 thousand to 7.40 million and 159.26 thousand. These paired descriptive differences show model-specific score–cost responses but do not establish statistical significance or a causal effect.

Figure 7 shows how these aggregate changes arise from task-level transitions. GPT-5.6-sol solves eight tasks only with scientific information and twelve only without it, whereas DeepSeek-V4-flash solves nine only with information and three only without it. Inspection of the GPT-5.6-sol cases suggests that scientific



(a) GPT-5.6-sol.

(b) DeepSeek-V4-flash.

Figure 7 Task-level overlap of Pass@1 success on the 91-task separable subset. (a) GPT-5.6-sol passes 29 tasks with scientific information and 33 tasks without it; 21 pass in both conditions, 8 only with information, and 12 only without it. (b) DeepSeek-V4-flash passes 21 tasks with scientific information and 15 tasks without it; 12 pass in both conditions, 9 only with information, and 3 only without it.

information can supply semantic constraints absent from local code symptoms, such as limiting cases, coordinate consistency, independent observables, and authoritative interfaces. However, it can also induce anchoring, scope spillover, or premature reliance on a supplied explanation, underscoring the need for executable evidence and independent validation.

Within this two-model comparison, scientific information produces a larger Pass@1 gain for the lower-baseline DeepSeek-V4-flash configuration than for GPT-5.6-sol. This pattern suggests that weaker-performing models may benefit more from external scientific guidance, whereas stronger models may rely less on it.

7 Limitations

The number of tasks in each scientific domain is still relatively limited, which may reduce the reliability of comparisons across domains. In addition, the analysis of the role of scientific knowledge in SWE tasks remains relatively preliminary, with insufficient exploration of how domain-specific knowledge is actually used and how to make it contribute to successful task completion.

8 Conclusion

We introduced **SWE-bench Science**, a repository-level benchmark for evaluating whether coding agents can repair scientific software while preserving its scientific contracts. The benchmark contains 119 tasks from 98 GitHub repositories across 20 scientific domains and organizes them into **Issue-driven**, **Expert-exploratory**, and **Engineering-integration** paradigms. Its Chain-of-Evidence Protocol combines separated public and private tests with metrics for repair progress, exact success, and regression preservation, making it possible to distinguish visible-test performance from complete private-test correctness.

Across eight coding-agent configurations, the strongest result is 47.90% Pass@1, while the same configuration attains a 96.64% public score. This gap, together with the four recurring scientific failure mechanisms identified in unsuccessful repairs, shows that repository-level scientific software engineering requires scientific abstraction, disciplined exploration, system-wide integration, and generalization beyond visible cases. Paired comparisons on the 91-task scientific-information subset further reveal model-dependent effects:

auxiliary scientific information lowers GPT-5.6-sol Pass@1 from 36.26% to 31.87% but raises DeepSeek-V4-flash from 16.48% to 23.08%, with opposite changes in token use. These findings indicate that supplying scientific information alone does not guarantee a better repair; its value depends on how effectively an agent connects that information to repository evidence and validates it through execution. SWE-bench Science provides a foundation for measuring this capability, while broader model coverage is needed to understand how scientific information affects agent performance on scientific software engineering tasks.

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A Scientific Domain Coverage

Table 4 Distribution of the 119 SWE-bench Science tasks across 20 scientific domains.

Scientific domain	Tasks
Chemistry	24
Materials Science and Engineering	16
Biology	13
Biomedical Engineering	12
Physics	11
Mathematics	7
Astronomy	7
Atmospheric Science	5
Civil Engineering	5
Surveying and Mapping Science and Technology	3
Geophysics	3
Mechanics	3
Electrical Engineering	2
Marine Science	2
Geography	1
Aeronautical and Astronautical Science and Technology	1
Nuclear Science and Technology	1
Computer Science and Technology	1
Statistics	1
Information and Communication Engineering	1
Total	**119**

B Task-Level Repository and Knowledge Inventory

This appendix provides the complete task-level annotation used to construct the benchmark. Each row links a task identifier to one of the 20 scientific domains used throughout the paper, together with its upstream repository, issue or pull request (when available), knowledge-domain scope, and scientific focus. The table contains all 119 tasks (IDs 001–119).

Table 5 Task-level scientific-domain, provenance, and scientific-knowledge inventory. A missing issue or pull-request reference indicates that no reference was provided in the source annotation.

Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
001	Chemistry	`Auto-Mech/autochem`	PR #637	TS linear segment cap selection, rotational symmetry number, reaction edge relabeling invariance, rotor observation	The rotational symmetry of the TS graph must be determined by the real linear segment and reaction edge topology; changing the atom number or writing it in reverse cannot change the rotor/symmetry observation.
002	Chemistry	`aditisingh4812/simplified_tuned_range_separated`	Not specified	range-separated tuning, local density structure, bohr-based grid scale and size consistency	In range-separated tuning, the units of the density cutoff are not converted together with the grid representation, and the local thresholds are therefore misaligned.
003	Materials Science and Engineering	`deepmodeling/dpdata`	Issue #973 / PR #983	stress/virial tag preservation, ASE compatibility, read and write consistency in extxyz/QUIP-GAP parsing	extxyz parsing must convert stress into virial and be compatible with tensor components and field aliases, otherwise the stress semantics will be distorted.

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Table 5 – continued from previous page

Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
004	Biology	`arq5x/bedtools2`	Issue #750	split-read exon overlap filtering, interval intersection semantics, hit determination on splicing boundaries	For split-read interval intersection, block overlap should be accumulated first and then fraction gate should be performed. Each block cannot be treated as an independent hit.
005	Biomedical Engineering	`schorschinho/osprey`	Issue #750 / PR #751	edited MRS subspectrum polarity correction, generalization from two subspectra to four subspectra, phase-cycle control propagation	Edited MRS requires polarity correction and phase-cycle control to be propagated through every subspectrum path; the fix cannot be confined to a single loader.
006	Biology	`arq5x/bedtools2`	Issue #954 / PR #959	Query database tail count after EOF, same-chrom / later-chrom tails, Fisher right tail p-value consistency	Fisher statistics require full database counts, and query EOF cannot truncate chromosome sweeps early.
007	Biomedical Engineering	`openmrslab/suspect`	Issue #33 / PR #34	spectral registration initial value symbol, warm-start convention drift, frequency shift and phase shift consistency	The drift sign of spectral registration must be consistent with the optimization residual agreement, otherwise the compensation direction will be reversed.
008	Physics	`PlasmaControl/DESC`	Not specified	first usable vacuum-plus-wall capability chain, outer-region operator payload, pseudo-vacuum metrics	In the vacuum-wall link, the outer-region geometry and coefficients must be transferred together, and different wall shapes cannot be pressed into the same diagnosis.
009	Physics	`PlasmaControl/DESC`	Not specified	projection-wall geometry construction, external-region path, downstream reduced payload propagation	The projection wall cannot degenerate into a scaled wall, and the boundary derivatives and geometric parameters must be entered into the outer-region calculation together.
010	Physics	`PlasmaControl/DESC`	Not specified	alternate perturbation-representation support, FOURIN, packed trig lookup, matrix assembly	The PARFAC-driven alternate perturbation representation must not fall back to the baseline; parity, phase, and coefficient assembly must all follow the alternate representation.
011	Physics	`PlasmaControl/DESC`	Not specified	response-gated sideband capability, localized exchange propagation, energy diagnostics	Sideband response must propagate through the support domain, coefficient assembly, and energy diagnostics; changing only a local cutoff is insufficient.
012	Physics	`PlasmaControl/DESC`	Not specified	3-D cross-spectral coupling, coefficient assembly, energy channel, mode-family generalization	3-D mode coupling must retain the effective support of the difference/sum two branches, and cross-modal coupling cannot be collapsed into a unified weight.
013	Physics	`PlasmaControl/DESC`	Not specified	free / soft-fixed / hard-fixed boundary regimes, operators and energy closure ladder	The boundary response should be closed to coefficients and energies respectively by interval and cannot be smoothed out by global averaging.
014	Physics	`PlasmaControl/DESC`	Not specified	prescribed wall geometry, vacuum-plus-wall chain, outer-region operator payload, reduced response blocks	The TERPSICHORE link must be closed end-to-end and cannot rely on a single fixture or plasma-only fallback.
015	Biomedical Engineering	`IlyaKuprov/Spinach`	Issue #22	spin-dynamics thermalization, formalism-specific identity/projector construction, unit-state representation consistency	thermalize should use formalism-specific identity/projector, and unit_state cannot be treated as two objects at the same time.
016	Biology	`macs3-project/MACS`	Issue #735 / PR #736	HMMRATAC semantics, blacklist filtering, query EOF post-count consistency	HMMRATAC refinement must first guard empty peak_content, and cannot close empty peaks on sparse bedGraph.
017	Astronomy	`gammapy/gammapy`	Issue #6754 / PR #6757	J-factor line-of-sight impact parameter, b = D sin(theta), r = b cosh(t) stable integration	The J-factor line integral must stabilize the branch according to the impact parameter, and cannot use tan(theta) to push directly to the singular point.
018	Biology	`macs3-project/MACS`	Issue #680 / PR #689	blacklist filtering, HMMRATAC training/inference path, peak call statistical consistency	The blacklist should be filtered before training and segmentation, and blacklist segments cannot be allowed to enter the model and counting of HMMRATAC.
019	Biology	`aertslab/pycisTopic`	PR #196	Discretization of topic probability matrix to region / cell set, legacy object API, Regions_in_binarized_topic maintenance	Topic binarization needs to support both matrix/object entries and stable threshold semantics, and cannot just cover the old object API.
020	Materials Science and Engineering	`deepmodeling/deepmd-kit`	PR #5581	neighbor graph, periodic-cell mapping, segment reductions, force / atom virial scatter	The neighbor graph needs to unify PBC mapping, empty segment reduction and force/virial scatter, and guard edges cannot be regarded as effective contributions.

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Table 5 – continued from previous page

Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
021	Chemistry	`deepmodeling/dpti`	PR #103	coupling endpoints, pair-specific epsilon, three-stage HTI/FEP input generation, thermo_style consumption	Soft-core LJ FEP must generate a real `compute fep`; energy divided by lambda cannot substitute for derivative observations.
022	Biomedical Engineering	`nilearn/nilearn`	Issue #5168 / PR #5169	Volume-to-surface atlas projection, discrete region identity preservation under nearest sampling, missing samples should return NaN	Discrete brain area projections should adopt mode sampling, and NaN should be retained for missing samples. Labels cannot be averaged numerically.
023	Chemistry	`MDAnalysis/mdanalysis`	Issue #4257 / PR #4258	NPT trajectory without jump expansion, box dimensions update, first frame displacement history initialization	NoJump's frame bookkeeping must preserve the first cross-boundary displacement and cannot exit early at frame 0.
024	Statistics	`pymc-devs/pymc`	Issue #6871 / PR #6872	ZeroSumNormal's log-density, normalization and pseudo-determinants on zero-sum manifolds	The logp of ZeroSumNormal must be calculated on the zero-sum manifold and cannot directly reuse ambient Gaussian normalization.
025	Biomedical Engineering	`schorschinho/osprey`	Issue #447 / PR #448; Issue #657 / PR #658	acquisition metadata consistency in the edited-MRS workflow, consistent calibration reports and representations	Edited-MRS acquisition metadata and the water-calibration time scale must remain consistent; TE/TR must not be converted a second time.
026	Atmospheric Science	`Unidata/MetPy`	Issue #3315 / PR #3324	Isentropic interpolation on sigma coordinates, pressure column selection, dimension semantic correction	Isentropic interpolation must respect vertical coordinate semantics such as sigma/eta, and cannot treat them directly as pressure columns.
027	Chemistry	`little1d/NMRTrans`	Issue #4	Masking and representation semantics of NMR peak-set encoding, batch padding, and set transformer	The mask semantics of the peak encoder are reversed, the padding is retained, the real peaks are masked, and the set representation is distorted.
028	Biology	`JialongJiang/DSPIN`	Not specified	control-referenced relative responses, perturbation response normalization and fallback path integrity	The control sample index and relative response baseline cannot be misaligned, otherwise the disturbance response will shift as a whole.
029	Biology	`biopython/biopython`	Issue #4611 / PR #4637	Feature coordinates and sequence semantics after circular record orientation normalization and reverse-complement	After circular feature reverse-complement, the part order must be reversed, and the original cross-origin order cannot be retained.
030	Materials Science and Engineering	`materialsproject/pymatgen`	Issue #3677 / PR #3779	SHELX/AIRSS.res export, SFAC ordered species column, stable round-trip serialization	The SFAC of.res is an ordered list, and species cannot be written as an unordered set.
031	Biomedical Engineering	`mne-tools/mne-python`	Issue #13913 / PR #13920	average reference across multiple sensor types, classification type re-reference and joint / projection path consistency	The average reference of mixed sensor types should be done separately according to channel type and cannot be directly combined into a union mean.
032	Chemistry	`openmm/openmm`	Issue #5025 / PR #5026	GROMACS topology import, NBFIX / combination rule semantics, total energy representation maintenance	GROMACS topology import needs to unify the translation semantics of NBFIX, combination rule and 1-4 exception.
033	Materials Science and Engineering	`pyscf/pyscf`	PR #2946	consistency between k-point and Gamma-supercell representations in periodic HF, Coulomb wrap-around semantics	Periodic HF k-point and Gamma-supercell paths must implement equivalent Coulomb semantics; wrap-around handling must not diverge between them.
034	Mathematics	`patrick-kidger/diffrax`	Issue #598 / PR #746	strong order, reverse-oriented SDE integration, boundary conditions and hidden state processing	The step size and Levy area parity of the reverse time SDE must be processed according to the direction, and the forward SRK convention cannot be followed.
035	Mathematics	`scipy/scipy`	Issue #14480 / PR #14552	Numerical consistency of LSODA event processing, dense output, stiff kinetics	The dense output of LSODA needs to take the history of the last successful step, and cannot use the next candidate state for event positioning.
036	Marine Science	`Parcels-code/Parcels`	Issue #2521 / PR #2598	Problems with spherical curvilinear grids, latitude and longitude grid index, and downstream parameters still follow the old path	Curved grid point-in-cell must be calculated according to spherical geometry and cannot continue to be approximated by longitude and latitude planes.
037	Biomedical Engineering	`nipy/nibabel`	PR #1340	Enhanced multiframe DICOM reconstruction, frame/direction semantics and image plane consistency	The frame order of multi-frame DICOM must be consistent with the physical slice order, and cannot be reconstructed solely by index.
038	Materials Science and Engineering	`spglib/spglib`	Issue #584 / PR #589	Classification stability, primitive/-conventional transformation and symmetry search under equivalent lattice representation	The operation route of symmetry search must be updated synchronously with the base transformation and cannot be disconnected from the structure route.

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Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
039	Geography	landlab/landlab	Issue #2408 / PR #2409	parcel transport, reach-scale flux consistency, segmentation and external representation consistency	The speed of parcel transport must be consistent with reach-level flux conversion, and the number of particles cannot be allowed to change the physical transport rate.
040	Geophysics	simpeg/simpeg	Issue #1099 / PR #1575	magnetic-dipole forward modeling, source representation equivalence, numerical accuracy and theoretical fit	Different formulations of magnetic dipole forward modeling must fall into the same discrete field space, and the representation method cannot change the results.
041	Geophysics	sunpy/sunpy	Issue #8406 / PR #8415	Earth-centered coordinate transformations, GEI/GSE path consistency and vector conservation	The GEI/GSE transformation must follow a purely rotational path and cannot go through an intermediate map with additional corrections.
042	Astronomy	sunpy/sunpy	Issue #8490 / PR #8548	Sun coordinate objects maintain frame, observer, obstime, representation, units and numerical meanings in table serialization	The solar coordinate table round-trip must retain frame, observer and obstime, and cannot compress the coordinate object into a pure numerical value.
043	Materials Science and Engineering	PMEAL/porespy	Issue #966 / PR #1174	Two-point correlation function, aperiodic boundaries, overlap counting and radial binning on finite images	Finite image two-point correlation needs to perform overlap and radial bin statistics according to finite-image semantics, and cannot be treated as a periodic domain.
044	Chemistry	pybamm-team/PyBaMM	PR #5600	In regularized power operations, distinguish between state-independent fitting coefficients and dynamic state-dependent expressions	The fractional-power stabilization in the exchange current expression must distinguish between constants and state variables, and the parameters themselves cannot be rewritten.
045	Marine Science	xgcm/xgcm	Issue #712 / PR #713	Connection surface direction, axis order, consistency of pad and transpose	cubed-sphere halo/padding must first align the dimensions and surface directions of the source slice, and cannot read the adjacent surfaces incorrectly after transpose.
046	Astronomy	StingraySoftware/stingray	PR #965	Coherence/power spectrum calculation under effective domain, boundary state and invalid value determination	The coherence estimate must correctly deduct the noise deviation and count as averaged spectra, and cannot return a finite value in the noise-dominated frequency band.
047	Information and Communication Engineering	scikit-rf/scikit-rf	Issue #654 / PR #655	Conversion and parameter consistency of different microwave network representations under complex reference impedance	Network connections under complex reference impedance must first undergo wave-definition transformation, and the traveling-wave convention cannot be used.
048	Chemistry	pycalphad/pycalphad	Issue #310 / PR #311	consistency between phase energy, Gibbs energy, and ordered-phase representation	Gibbs-energy bookkeeping for ordered/disordered phases must project onto the non-ordering sublattice; database encoding must not change phase stability.
049	Mathematics	zwicker-group/py-pde	PR #839	Multi-step physical process semantics and incombinability in the evolution of discrete random fields	The noise terms and volume factors of discrete random fields must be processed hierarchically, and cell-volume scaling cannot be placed in the wrong location.
050	Physics	jcmgray/quimb	Issue #402	Scale-dependent norm reconstruction, result consistency of different tensor representations	The norm of the tensor network must remain unchanged after gauging/rescaling, and the approximate contraction cannot change the exact norm.
051	Geophysics	SHTOOLS/SHTOOLS	PR #397	Spherical harmonic gradient, polar boundary and planetary magnetic field calculation	The gravity/magnetic force second-order tensor must be consistent from the local frame to the Cartesian frame, and recursion and boundary constraints cannot be allowed to destroy the rotation.
052	Materials Science and Engineering	usnistgov/fipy	Issue #1190 / PR #1192	Modular gradients and physical geometry consistency on non-uniform orthogonal grids	Non-uniform Grid3D needs to unify cell widths, face sizes and modular gradient semantics, and cannot only retain a single spacing.
053	Biomedical Engineering	Project-MONAI/MONAI	Issue #4103 / PR #6681	Physical spacing, subvoxel surface area approximation and surface Dice	Surface Dice should be calculated based on physical surface area, and anisotropic voxels cannot simply be regarded as equal-area boundaries.

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Table 5 – continued from previous page

Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
054	Chemistry	matchms/matchms	PR #863	Screening, merging and entropy-based spectral similarity of highly similar peaks	Entropy similarity cannot allow overlapping candidate windows to count the same peak repeatedly, and the pair/-matrix paths must share the same peak semantics.
055	Astronomy	astropy/specutils	Issues #817 / #1059 / #1300; PRs #1060 / #1305	Boundary extension, partially overlapping bins and finite interval filling	Spectral resampling must propagate flux and variance according to limited bin overlap, and bins cannot be used as point samples.
056	Aeronautical and Astronautical Science and Technology	poliastro/poliastro	Issue #1563; PR #1583	Consistency between orbital state and binding epoch, history sampling semantics	Each state and epoch of Orbit->Ephem must point to the same orbital state, and the anomaly cannot drift with the timeline.
057	Chemistry	mdtraj/mdtraj	Issue #1925; PR #1929	Molecular reconstruction under periodic boundaries, bonded geometry and coordinate order independence	make_molecules_whole restores the periodic image along the bond graph in one pass, and cannot let the bond order determine the translation results.
058	Nuclear Science and Technology	openmc-dev/openmc	Issue #3852; PR #3853	Rectangular lattice distance, boundary and geometric determination in particle transport	The boundary distance and transport event of the repeated rectangular lattice must be calculated correctly, and the legal geometry cannot be misjudged as terminated.
059	Computer Science and Technology	scikit-image/scikit-image	Issue #5112; PR #6296	voxel spacing, area and morphometric calibration of regionprops	regionprops should treat spacing as a physical scale and cannot treat anisotropic sampling as a lattice index.
060	Biomedical Engineering	SpikeInterface/spikeinterface	Issue #3328; PR #3517	non-equidistant sampling, cross-band motion correction, and interpolation correction	Motion interpolation must align AP/LF streams on the physical time axis; assuming uniform motion from the sampling rate is invalid.
061	Astronomy	astropy/reproject	Issue #199; PR #589	Numerical elimination, reprojection and energy conservation of small sphere area	The extremely small sky pixel in high-resolution reproject will have numerical cancellation, and overlap geometry and flux conservation must be stable.
062	Materials Science and Engineering	materialsproject/atomate2	Issue #333 / PR #415	Green-Lagrange strain-stress fitting, independent shear response reconstruction, and tensor symmetry in a high-throughput crystal elasticity workflow	Elastic tensor reconstruction must bring the complete shear response back to the final result, and the reduce/reconstruct link cannot lose symmetry information.
063	Biology	tskit-dev/tskit	Issue #3053, #3175 / PR #3059, #3176	Distinguishing between "no lineage" and "multi-rooted trees with no common ancestors" under window normalization, and normalizing the observation span	Window normalized pair-coalescence needs to distinguish between non-lineage trees and multi-rooted trees without common ancestors, and the boundary semantics of the two types cannot be confused.
064	Chemistry	cclib/cclib	Issue #660 / PR #745	Signed energy extraction, unit consistency and ccData exposure for DFT-D dispersion correction in ORCA output	ORCA's DFT-D dispersion correction should be written out as an independent signed physical quantity and not just left inside the parser.
065	Astronomy	yt-project/yt	Issue #4788 / PR #4939	SPH-based three-dimensional projection, oblique slicing, real sight integral and kernel function geometric semantics	SPH line-of-sight integration and oblique slicing need to share consistent kernel geometry and normalization, and the resolution cannot be allowed to change the observation results.
066	Atmospheric Science	SciTools/iris	Issue #4830 / PR #5548	Equivalent identification of Rotated Mercator and Oblique Mercator under CF grid mapping, projection object and site matching semantics	The CF grid mapping of rotated Mercator and oblique Mercator should be recognized as equivalent and cannot be linked to the wrong projection reference.
067	Atmospheric Science	Ouranosinc/xclim	Issue #700 / PR #704	Overwintering precipitation and Drought Code state transfer, recursive boundary between active season and dormant phase when seasons switch	The seasonal state machine of the fire danger index should separate the winter precipitation and active-day updates, and the state transfer cannot be connected to the wrong stage.
068	Biology	samtools/htslib	Issue #1820 / PR #1897	Actual coverage interval, rlen, END / symbolic allele / reference block semantics recorded on the reference genome	The reference occupancy interval of VCF/BCF should be consistent on paths such as symbolic allele, END and SVLEN.

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Table 5 – continued from previous page

Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
069	Biology	open2c/cooltools	Issue #456 / PR #501	Contact-frequency and smoothed P(s) propagation, normalization and multi-region aggregation consistency	expected cis smoothing needs to propagate the numerators and denominators of raw contact and weighted contact separately, and observation apertures cannot be mixed.
070	Biology	levitsky/pyteomics	Issue #202 / PR #203	ProForma local modification and global ion-carrying charge calculation, charge_state consistency	ProForma's global loads, local modifications, and neutral masses/m/z require unified charge accounting, and electrons cannot be counted repeatedly.
071	Physics	silx-kit/pyFAI	Issue #2773 / PR #2772	Transformation from detector direction to sample coordinate system q vector, exit-angle and reciprocal represent consistency	The q vector transformation must maintain the consistency of the detector to sample coordinates, and the orientation encoding cannot change the reciprocal space results.
072	Biomedical Engineering	pydicom/pydicom	Issue #2210 / PR #2212	Bit depth, sampling, limited-range luma/chroma conversion of YBR_FULL / YBR_FULL_422 / YBR_PARTIAL_420	The YBR_* pixel format must unify bit depth and sampling extension semantics, and cannot just modify the limited-range path.
073	Biology	biotite-dev/biotite	Issue #777 / PR #781	Conformational connectivity, bond map preservation and bond mapping after alternate location selection	The alternate location filtering must be completed before the bond graph, and the selected conformation cannot be disconnected from the struct_conn.
074	Materials Science and Engineering	deepmodeling/DeePTB	Issue #332 / PR #320	Generalized eigenvalue problems, ill-conditioned overlap matrix processing, finite band data and validity masks	The generalized eigenvalue problem of non-orthogonal tight binding must first deal with the ill-posed overlap matrix, and cannot directly Cholesky on the invalid subspace.
075	Materials Science and Engineering	deepmodeling/deepmd-kit	Issue #5593 / PR #5601	Boundary interpolation, first-order sensitivity, layout and derivative consistency of DeePMD compressed tables	Compressed table extrapolation must maintain both value and derivative continuation semantics, and C1 continuation cannot be truncated outside the boundary.
076	Mathematics	cvxpy/cvxpy	Issue #3092 / PR #3093	Coordinate mapping of N-dimensional parameters to sparse coefficient matrix after indexing, broadcasting, and matrix multiplication	Canonicalization of the Parameter tensor requires first folding repeated positions and then mapping to a sparse coefficient matrix.
077	Mathematics	shapely/shapely	Issue #1670, #1929 / PR #1708, #1939	Numerical stability fallback, candidate edge enumeration, NumPy ufunc batching and output buffering compatibility	Oriented envelope must truly select the smallest area among candidate edges and is compatible with NumPy batch processing.
078	Astronomy	astropy/photutils	Issue #2287 / PR #2288	Bidirectional transformation, direction consistency and round-trip constraints of aperture geometry under WCS	The conversion from sky aperture to pixel aperture must keep the WCS local base direction consistent, and the aperture mirror cannot be shifted.
079	Chemistry	aspuru-guzik-group/selfies	Issue #120 / PR #125	kekulization of SMILES aromatic system to SELFIES, charged aromatic carbon electron counting	When converting the SMILES aromatic system to SELFIES, the electron count of the charged aromatic carbon must also be dealt with, and the neutral zero-hydrogen path cannot only be repaired.
080	Surveying and Mapping Science and Technology	xarray-contrib/xarray-spatial	Issue #1086 / PR #1087	footprint, coordinate direction normalization, nodata propagation and cubic boundary interpolation	The footprint, coordinate direction and nodata propagation during reprojection must be bound to the interpolation order, and the boundaries cannot be allowed to degenerate into copied values.
081	Biology	scverse/anndata	Issue #1854 / PR #2122	backed HDF5/Zarr re-indexing, outer join, and sparse/dense annotation consistency	Disk-backed concatenation must write non-concatenated-axis annotations back under the new index; preserving only the main matrix is insufficient.
082	Materials Science and Engineering	deepmodeling/dpgen	Issue #1639 / PR #1801	DP-GEN active-learning input generation, LAMMPS template revise, model-deviation control items, relative error and electron temperature propagation	DP-GEN must propagate the same set of LAMMPS model-deviation settings in both direct generation and template revision paths, and the relative, cluster or electron temperature items cannot be lost in the template path.

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Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
083	Chemistry	deepmodeling/reacnetgenerator	Issue #2165 / PR #2221	bond-order trajectory, atom-bond graph equivalence, isomer merge mode, cyclic structure species naming and reaction network report	The species report of ReacNetGenerator must define the same molecule according to chemical graph equivalence and merge mode, and cannot separate bond order changes or traversal order into different species.
084	Materials Science and Engineering	deepmodeling/dpdata	Issue #951 / PR #957	ABACUS relaxation-output parsing, frame-wise coordinates and cell labels, archive/reload round-trip, complete trajectory recovery	The ABACUS relaxation round-trip must preserve the complete relaxation history and frame-wise labels, rather than retaining only the final structure or losing cell/coordinate semantics.
085	Chemistry	dftbplus/dftbplus	Issue #1069 / PR #1462	DFTB+ COSMO/GB archived observations, atomic order invariance, equivalent atomic records and solvent surface adjacencies	DFTB+ implicit solvent observations should be insensitive to equivalent atoms and input order, and should not allow the order of atom records to change COSMO-like reporting.
086	Materials Science and Engineering	materialyzeai/matgl	Issue #642 / PR #819	MatGL dataset element list, pre-trained model element vocabulary, graph construction atom type encoding, fine-tuning compatibility check	MatGL fine-tuning must first verify that the data set element list is consistent with the model element vocabulary, and the graph construction cannot be allowed to silently map atoms to wrong embeddings.
087	Biomedical Engineering	dipy/dipy	Issue #3968 / PR #3976	Patch2Self small batch denoising, trailing-volume flush, b0 volume data processing, patch radius and baseline shift	Patch2Self wants to keep the diffuse volume consistent with b0 semantics on mini-batch and tail volume data, and cannot leave finite but offset denoised observations.
088	Atmospheric Science	pydata/xarray	Issue #9108 / PR #9116	CFTimeIndex resample, calendar and has_year_zero reserved, year-zero near month/year-end boundary aggregation	CF time aggregation must retain calendar and has_year_zero semantics, and cannot aggregate legal short sequences into failure paths at the year-zero boundary.
089	Chemistry	cclib/cclib	Issue #46 / PR #1085	ORCA excited-state absorption spectrum, oscillator strength, multiple transition attribute table, etoscs representation selection	cclib must retain the correct oscillator-strength representation when parsing ORCA excited state spectra, and must not allow subsequent or substitute spectra to overwrite semantic values.
090	Chemistry	jjhelmus/nmrglue	Issue #187 / PR #189	Bruker NUS compressed FID stream, nuslist sparse coordinates, quadrature/acquisition order, missing FID zero-fill	nmrglue needs to expand the sparse FID stream into a multi-dimensional NMR grid according to the nuslist and collection order, and cannot treat the compressed stream as an ordinary dense 2D sequence.
091	Materials Science and Engineering	materialsproject/pymatgen-core	Issue #4607 / PR #87	Gaussian cube nonzero origin, Bohr-to-angstrom unit conversion, voxel origin and structural coordinate placement	When reading Gaussian cube, the nonzero volume data origin and unit conversion must be included in the structure placement. The first voxel cannot be the default coordinate origin.
092	Chemistry	chemprop/chemprop	Issue #426 / PR #427	Condensed Graph of Reaction, atom mapping, reaction-mode options, and reactant/product feature propagation	Chemprop reaction featurization must construct a CGR according to the reaction mode and atom map; mapped atoms must be retained and reactant/product changes must not be encoded in reverse.
093	Biomedical Engineering	Project-MONAI/MONAI	Issue #6985 / PR #7000	ITKReader, MetaTensor, ITK image round-trip, affine/spacing/origin/direction coordinate convention	MONAI's reader -> MetaTensor -> ITK round-trip must preserve patient-space coordinates and orientation, image geometry cannot be silently changed between equivalent paths.
094	Physics	materialsproject/pymatgen	PR #1706	magnetic space group operations, BNS/Jones-Faithful setting transform, fractional translation, time reversal	Magnetic space group archive records must be converted according to setting to preserve translation and time-reversal semantics, and the old coordinate convention cannot be read directly as the current convention.
095	Materials Science and Engineering	PMEAL/OpenPNM	Issue #2843 / PR #2849	throat conductance field, model builder, operator assembly, advection-diffusion solver, repeated stage transfer	Porous network transmission requires consistent transmission of conductance representation from model output to solver and flux calculations. Object fields and array fields cannot be mixed in repeated solves.

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Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
096	Atmospheric Science	CLIMADA-project/ climada_python	Issue #831 / PR #839	antimeridian longitude normalization, track bounds, centroid filtering, sparse hazard matrix, equivalent date-line encoding	Tropical cyclone hazard calculations need to unify longitude wrapper semantics in paths, centroid ranges, and sparse maps, and cannot miss or misplace wind fields near ±180 degrees.
097	Physics	deepmodeling/ deepmd-kit	Issue #5663 / PR #5854	Spin virtual types, negative placeholder atoms, dense/lower layouts, dpmodel/PT parity, padded mixed-size batches	DeePMD spin model preprocessing requires negative type padding to remain inert in real/virtual type expansion and backend layout, and cannot change placeholder atoms into valid spin types.
098	Surveying and Mapping Science and Technology	pytroll/satpy	Issue #3182 / PR #3217	geostationary satellite position units, TLE fallback, cloud-top-height parallax, observer-look boundary, inverse resampling	Satpy parallax correction needs to unify internal satellite height/radius units, only convert at external library boundaries, and cannot mix meters/km between reader metadata, TLE and Cartesian geometry.
099	Surveying and Mapping Science and Technology	Stanford-NavLab/ gnss_lib_py	PR #121	WLS pseudorange positioning, Earth-rotation/Sagnac correction, satellite ECEF convention, simulator direct satellite states	GNSS WLS point positioning requires rotating the satellite position to the receive measurement frame and being compatible with simulator input. Transmitted-frame ECEF cannot be directly solved as receive-frame.
100	Chemistry	DCoupry/ autografs	Issue #174/#197/#198/#207 / PR #208	AuToGraFS relaxed embedding, topology slot motion, closure regression, cell volume scaling, pinned-net control	Relaxed framework embedding must allow low symmetry slot movement while maintaining topology, closure and volume scale, the objective cannot be adjusted to allow individual nets to pass.
101	Chemistry	pybamm-team/ PyBaMM	Issue #5414 / PR #5540	half-cell Doyle-Fuller-Newman model, surface-form electrolyte potential, boundary-condition registration, thermal Ohmic-heating coupling	PyBaMM half-cell DFN must consistently register electrolyte/solid phase potential boundary conditions for surface-form and thermal sub-models, and cannot allow half-cell domain splitting to cause gradient shape mismatch.
102	Materials Science and Engineering	marrink-lab/ polyply_1.0	Issue #380 / PR #381	polymer residue graph, template/volume identity, coordinate generation, library loading, residue equivalence	Polymer workflows must pass template, volume, and residue graph identities all the way to coordinate generation without losing graph structure semantics between loading and generation.
103	Mechanics	SteveDoyle2/ pyNastran	Issue #584 / PR #85	symmetric PCOMP expansion, ABD equivalent, D block calculation, explicit layup and symmetric layup consistency	The ABD of the symmetrical layup must be consistent with the explicit layup, and the expanded directional stiffness and D matrix cannot deviate.
104	Chemistry	pybamm-team/ PyBaMM	Issue #3526 / PR #3979	upwind/downwind edge states, cell-centered and edge-centered flux, Dirichlet FV boundary handling, mesh-size invariance	The three-layer semantics of cell/edge/boundary in a finite volume must be consistent, and upwind/downwind cannot be mismatched in boundary processing.
105	Chemistry	rdkit/rdkit	Issue #9338 / PR #9339	tautomer canonicalization, stereo preservation, V3000 structure serialization, atropisomer bond validity	Tautomeric normalization must preserve chirality and output serializable molecules, and the stereochemistry cannot be broken in canonicalize.
106	Chemistry	MLCIL/ scikit-fingerprints	Issue #519 / PR #525 / PR #528	MAP4 / MAPC retrieval protocol, chirality invariance, representation consistency, benchmark panel, version pinning	Molecular retrieval protocols need to maintain the same semantics across chirality, representation, and batch processing paths, and cannot be true only for visible panels.
107	Mathematics	sympy/sympy	Issue #28186 / PR #28189	rational integration, exact root handling, log_to_real, derivative identity, real antiderivative semantics	The original function returned by rational integration must really be able to convert back to the original integrand, and root processing cannot break the derivative identity.
108	Mathematics	mpmath/mpmath	Issue #1104 / PR #1107; Issue #930 / PR #1111	theta3/theta4 fixed-precision paths, near-unit nome stability, tiny-nome cancellation, derivative propagation, modular transform	Jacobi theta must also be stable on legal nomes. It cannot be rejected by mistake, nor can it explode on the path of small nomes.
109	Mechanics	elhajjar1/ wrinkleFE	Issue #252 / PR #285	multi-wrinkle field composition, fiber-angle propagation, phase offsets, FE analysis, RSS shortcut	The multi-wrinkle field cannot be approximated by single wrinkle RSS, and the fiber angle must follow the synthesized wrinkle field.

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Task	Scientific domain	Upstream repository	Issue/PR	Knowledge scope	Task focus
110	Civil Engineering	robbievanleeuwen/section-properties	Issue #272 / PR #287	first-moment result, outline quantity semantics, cross-section property consistency, reported vs constructed geometry	The first-order moment result cannot be mistakenly reported as an outline quantity, and the report semantics and construction geometry must be consistent.
111	Mechanics	GazzolaLab/PyElastica	Issue #81 / PR #87	tapered slender filaments, inverse-time material-response, cross-section variation, translational observations, free-response consistency	The free response of the slender body cannot jump randomly with the change of cross section, and the material response and geometric changes must be coordinated.
112	Civil Engineering	robbievanleeuwen/section-properties	Issue #460 / PR #462	hybrid-section plastic capacity, constituent force/centroid preservation, split geometry pieces, plastic neutral axis	The plastic load-bearing of the composite section must preserve the force and centroid of each material segment, and cannot just report a seemingly reasonable number.
113	Electrical Engineering	pvlib/pvlib-python	Issue #2054 / PR #2055	timezone-aware timestamps, calendar bins, day-of-year, extraterrestrial irradiance, SPA, equivalent aware representations	Solar resource calculations must give consistent results for "different time representations of the same moment", and time zone normalization cannot be aliased.
114	Electrical Engineering	PyPSA/PyPSA	Issue #1360 / PR #1371	storage history, planning-epoch boundary, cyclic vs initial state, Store/StorageUnit boundary semantics, state recurrence	The energy storage state must abide by the cycle boundary priority rules before and after the planning boundary, and the initial state cannot be mistaken for the previous period.
115	Civil Engineering	JWock82/Pynite	Issue #255, #275 / PR #323	Global stiffness singularity, structural stability determination, sparse/dense solution consistency, zero right end boundary	Legal frames cannot return finite but distorted displacements when unstable. Global stiffness singularities, zero load boundaries, and sparse/dense path consistency must be determined together.
116	Civil Engineering	JWock82/Pynite	Issue #255, #275 / PR #323	Structural response reports, equilibrium equations, matrix storage modes, physical interpretation of control studies	Response reporting by the same framework in sparse / dense matrix storage modes must remain physically interpretable and consistent with equilibrium equations and API conventions.
117	Civil Engineering	JWock82/Pynite	Issue #317 / PR #319	inactive member results, node displacement chord, physical member segmentation, local/global coordinate transformation	The inactive member still needs to be linearly interpolated along the chord after displacement to ensure that the scalar / sampled results are compatible with the geometry.
118	Chemistry	tblite/tblite	PR #326 / PR #343	GFN2-xTB multipole electrostatics, 3-D periodic Ewald decomposition, energy/force/stress consistency, unit cell translation and derivative cache	Different legitimate periodic unit cell representations of the same molecule should give consistent energies, forces, and stresses; multipolar Ewald's real/reciprocal paths and cached derivatives must be aligned.
119	Chemistry	MolSSI/QCEngine	PR #349	Turbomole nprhessian/hessian selection, 3N×3N Cartesian Hessian, projected/unprojected curvature, QCSchema result consistency	Turbomole Hessian analysis must recognize nprhessian/hessian records, infer the full 3N×3N dimensions, and pass the unprojected Cartesian curvature to downstream reaction path/vibration analysis.